

Title: Digital Thread Readiness for Shipbuilding — Concept Deck

Audience: Engineering director, IT/PLM sponsor, program manager

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Note: Portfolio outline; full visual deck built via `plm-slide-builder` skill and PowerPoint COM.

Slide 1 — Title

Digital Thread Readiness for Shipbuilding

From design release to production and class evidence

Prepared for: [Anonymized shipyard]

Joonha Lee | PLM & System Integration

Slide 2 — Executive thesis

- Shipyards lose schedule and rework cost when engineering data, change impact, and evidence are fragmented.

- A governed digital thread connects authoring tools, PLM objects, change control, and verification artifacts.

- Goal: one traceable path from requirement → design → release → manufacturing → evidence.

Slide 3 — Current state (typical pain)

Area	Symptom	Business impact
BOM / configuration	Multiple Excel sources	Wrong material to block
Change control	Email + offline redlines	Late impact discovery
Drawings	CAD vault vs PLM mismatch	Rework at production
Evidence	Scattered PDFs	Class / audit risk

Slide 4 — Target operating model

- Authoring: AVEVA E3D/Marine/Draw on Windows workstations

- Control plane: API-based PLM (objects, structures, ECR/ECO, approvals) on Linux cluster

- Bridge: Local Agent / Plugin callbacks — not direct CAD-on-Linux

- Cloud: CONNECT for intelligence and optional AI assistant (governed)

Slide 5 — Digital thread data objects

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Requirement → Design object → E-BOM/MTO → Baseline → ECR → ECO → M-BOM → Evidence pack

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Key gates: AI/rule gate at promote steps; effectivity (hull/block/option/date); approval workflow.

Slide 6 — Vendor landscape (summary)

Capability	AVEVA	Siemens TC	Dassault 3DX	Hexagon
Marine 3D authoring	Strong	Partner	Partner	Niche
Unified engineering	Strong	Strong	Strong	Moderate
Open API / custom PLM layer	Flexible	Mature	Mature	Varies
Shipyard references	High (target yard)	High	High	Selective

Full benchmark available as separate deliverable.

Slide 7 — Phased roadmap (MVP)

Phase	Duration	Outcome
Phase 0 — Discovery	4 wks	Process map, data objects, gap list
Phase 1 — MVP thread	12 wks	One hull class: req → ECO → evidence
Phase 2 — Scale	6 mo	Multi-discipline, ERP/MES touchpoints
Phase 3 — Optimize	Ongoing	KPI dashboard, AI-assisted checks

Slide 8 — KPI model

KPI	Definition	Gate
Release readiness	% objects with complete evidence	≥ 95% for production release
Change impact coverage	ECR items with linked affected objects	100% for Class 1 changes
BOM accuracy	M-BOM match to released E-BOM	Zero critical mismatches
Cycle time	Req → approved release	Target –20% vs baseline

Slide 9 — Risks & mitigations

Risk	Likelihood	Mitigation
Legacy data migration	High	Phased object scope; cleansing sprint
Authoring vs PLM split	Medium	Hybrid architecture clarity; agent bridge
User adoption	Medium	Champion network; template libraries
Integration overload	Medium	MVP boundary; API-first interfaces

Slide 10 — Recommended next step

10-page concept alignment workshop + MVP charter (6 weeks)

Deliverables:

- Validated process map for one pilot block/hull
- Object model and integration sketch
- Executive one-pager for funding decision

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Build instructions (internal)

To produce full PPTX from this outline:

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```powershell
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Use existing PLM builders — copy canonical deck first

**\$src = "E:\이력서\AVEVA - Marine Principal Technical Support & Consultant - PLM SME, Busan\Proposal\Future\_Industrial\_PLM\_Meeting\_Deck\_EN.pptx"**

Or build fresh 10-slide deck via plm\_slide\_work\build\_summary\_en.ps1 pattern

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Reference memory: ``.cursor/memory/aveva-plm-application-deck.md``